Air System Sizing Summary for DOAS-1-CORRIDORS ONLY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:25 PM

Air System Information

Air System Name	**DOAS-1-CORRIDORS ONLY**		Number of zones	**1**	
Equipment Class	**PKG ROOF**		Floor Area	**12123.5**	sqft
Air System Type	**MUA/DOAS**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Cooling Coil Sizing Data

Total coil load	**1.9**	Tons	Load occurs at	**July 15:00**	
Total coil load	**22.7**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**22.7**	MBH	Entering DB / WB	**106.1 / 66.2**	F
Coil CFM at July 15:00	**708**	CFM	Leaving DB / WB	**73.5 / 55.4**	F
Max coil CFM	**708**	CFM			
Sensible heat ratio	**1.000**				
Water flow @ 10.0 F rise	**N/A**				

Heating Coil Sizing Data

Max coil load	**33.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**708**	CFM	Ent. DB / Lvg DB	**20.0 / 68.5**	F
Max coil CFM	**708**	CFM			
Water flow @ 20.0 F drop	**N/A**				

Ventilation Fan Sizing Data

Design CFM	**708**	CFM	Fan motor BHP	**0.39**	BHP
Design CFM/sqft	**0.06**	CFM/sqft	Fan motor kW	**0.31**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**708**	CFM	CFM/person	**143.99**	CFM/person
CFM/sqft	**0.06**	CFM/sqft			

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Air System Information

Air System Name .... **DOAS-1-CORRIDORS ONLY**
Equipment Class .... **PKG ROOF**
Air System Type .... **MUA/DOAS**

Number of zones **1**
Floor Area .... **12123.5** sqft
Location .... **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months .... **Jan to Dec**
Sizing Data .... **Calculated**

Zone CFM Sizing .... **Sum of space airflow rates**
Space CFM Sizing .... **Individual peak space loads**

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
DOAS-1-CORRIDORS ONLY	58.1	July 17:00	27.3	12123.5

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
DOAS-1-CORRIDORS ONLY						
101-CORRIDOR (LOBBY)	1.2	June 19:00	25	0.4	413.0	0.06
108-GUEST CORRIDOR	2.1	June 10:00	12	1.7	195.2	0.06
109-GUEST CORRIDOR	2.2	July 17:00	20	0.4	336.5	0.06
113-RESTROOM	0.1	September 16:00	0	0.0	72.2	0.00
114-RESTROOM	0.2	September 16:00	0	0.0	73.1	0.00
115-SITTING AREA	2.0	July 18:00	26	0.2	185.5	0.14
118-ELEV CORRIDOR	4.0	July 17:00	33	0.7	556.4	0.06
137-GUEST CORRIDOR	2.7	June 18:00	26	1.7	433.0	0.06
138-POOL CORRIDOR	0.9	July 17:00	10	1.3	164.1	0.06
139-RESTROOM	0.1	July 20:00	0	0.0	74.1	0.00
140-RESTROOM	0.3	July 18:00	0	0.6	74.5	0.00
201-GUEST CORRIDOR	3.2	June 9:00	50	1.6	825.8	0.06
202-MECH/IDF	2.0	July 21:00	0	0.0	108.7	0.00
203-ICE	0.1	July 22:00	5	0.0	77.7	0.06
204-ELEV LOBBY	0.4	July 21:00	16	0.1	265.4	0.06
207-GUEST CORRIDOR	4.2	August 17:00	53	2.1	886.2	0.06
208-HOUSEKEEPING	1.5	August 13:00	24	0.2	239.2	0.10
301-GUEST CORRIDOR	3.2	June 10:00	51	1.5	844.0	0.06
302-MECH/IDF	2.1	July 20:00	0	0.0	112.9	0.00
303-ICE	0.1	July 22:00	5	0.0	82.7	0.06
304-ELEV LOBBY	0.4	August 16:00	16	0.0	259.7	0.06
307-GUEST CORRIDOR	4.1	August 17:00	50	1.9	832.0	0.06
308-HOUSEKEEPING	0.6	August 13:00	15	0.2	249.1	0.06
401-GUEST CORRIDOR	3.2	June 10:00	51	1.6	842.6	0.06
402-MECH/IDF	2.0	July 20:00	0	0.0	110.5	0.00
403-ICE	0.1	July 22:00	5	0.0	83.0	0.06
404-ELEV LOBBY	0.4	July 20:00	16	0.1	263.7	0.06
407-GUEST CORRIDOR	4.1	August 17:00	49	2.0	819.6	0.06
408-HOUSEKEEPING	0.6	August 13:00	15	0.2	250.9	0.06
501-GUEST CORRIDOR	3.9	July 17:00	51	3.2	847.9	0.06
502-MECH/IDF	2.4	July 17:00	0	0.2	115.0	0.00
503-ICE	0.2	July 18:00	5	0.2	81.2	0.06
504-ELEV LOBBY	0.9	July 18:00	16	0.6	259.9	0.06
507-GUEST CORRIDOR	5.9	July 17:00	50	3.6	840.9	0.06
508-HOUSEKPEEING	1.0	July 17:00	15	0.7	247.3	0.06

1. Summary

Ventilation Sizing Method .... **Sum of Space OA Airflows**
Design Ventilation Airflow Rate .... **708** CFM

2. Space Ventilation Analysis

Zone Name / Space Name	Floor Area (sqft)	Maximum Occupants	Maximum Supply Air (CFM)	Required Outdoor Air (CFM/person)	Required Outdoor Air (CFM/sqft)	Required Outdoor Air (CFM)	Required Outdoor Air (% of supply)	Required Outdoor Air (ACH)	Uncorrected Outdoor Air (CFM)	Direct Exhaust Air (CFM)
DOAS-1-CORRIDORS ONLY										
101-CORRIDOR (LOBBY)	413.0	0.0	24.8	0.00	0.06	0.0	0.0	0.00	24.8	0.0
108-GUEST CORRIDOR	195.2	0.0	11.7	0.00	0.06	0.0	0.0	0.00	11.7	0.0
109-GUEST CORRIDOR	336.5	0.0	20.2	0.00	0.06	0.0	0.0	0.00	20.2	0.0
113-RESTROOM	72.2	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
114-RESTROOM	73.1	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
115-SITTING AREA	185.5	3.0	26.1	5.00	0.06	0.0	0.0	0.00	26.1	0.0
118-ELEV CORRIDOR	556.4	0.0	33.4	0.00	0.06	0.0	0.0	0.00	33.4	0.0
137-GUEST CORRIDOR	433.0	0.0	26.0	0.00	0.06	0.0	0.0	0.00	26.0	0.0
138-POOL CORRIDOR	164.1	0.0	9.8	0.00	0.06	0.0	0.0	0.00	9.8	0.0
139-RESTROOM	74.1	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
140-RESTROOM	74.5	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	75.0
201-GUEST CORRIDOR	825.8	0.0	49.5	0.00	0.06	0.0	0.0	0.00	49.5	0.0
202-MECH/IDF	108.7	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
203-ICE	77.7	0.0	4.7	0.00	0.06	0.0	0.0	0.00	4.7	0.0
204-ELEV LOBBY	265.4	0.0	15.9	0.00	0.06	0.0	0.0	0.00	15.9	0.0
207-GUEST CORRIDOR	886.2	0.0	53.2	0.00	0.06	0.0	0.0	0.00	53.2	0.0
208-HOUSEKEEPING	239.2	1.9	23.9	5.00	0.06	0.0	0.0	0.00	23.9	0.0
301-GUEST CORRIDOR	844.0	0.0	50.6	0.00	0.06	0.0	0.0	0.00	50.6	0.0
302-MECH/IDF	112.9	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
303-ICE	82.7	0.0	5.0	0.00	0.06	0.0	0.0	0.00	5.0	0.0
304-ELEV LOBBY	259.7	0.0	15.6	0.00	0.06	0.0	0.0	0.00	15.6	0.0
307-GUEST CORRIDOR	832.0	0.0	49.9	0.00	0.06	0.0	0.0	0.00	49.9	0.0
308-HOUSEKEEPING	249.1	0.0	14.9	5.00	0.06	0.0	0.0	0.00	14.9	0.0
401-GUEST CORRIDOR	842.6	0.0	50.6	0.00	0.06	0.0	0.0	0.00	50.6	0.0
402-MECH/IDF	110.5	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
403-ICE	83.0	0.0	5.0	0.00	0.06	0.0	0.0	0.00	5.0	0.0
404-ELEV LOBBY	263.7	0.0	15.8	0.00	0.06	0.0	0.0	0.00	15.8	0.0
407-GUEST CORRIDOR	819.6	0.0	49.2	0.00	0.06	0.0	0.0	0.00	49.2	0.0
408-HOUSEKEEPING	250.9	0.0	15.1	5.00	0.06	0.0	0.0	0.00	15.1	0.0
501-GUEST CORRIDOR	847.9	0.0	50.9	0.00	0.06	0.0	0.0	0.00	50.9	0.0
502-MECH/IDF	115.0	0.0	0.0	0.00	0.00	0.0	0.0	0.00	0.0	0.0
503-ICE	81.2	0.0	4.9	0.00	0.06	0.0	0.0	0.00	4.9	0.0
504-ELEV LOBBY	259.9	0.0	15.6	0.00	0.06	0.0	0.0	0.00	15.6	0.0
507-GUEST CORRIDOR	840.9	0.0	50.5	0.00	0.06	0.0	0.0	0.00	50.5	0.0
508-HOUSEKPEEING	247.3	0.0	14.8	5.00	0.06	0.0	0.0	0.00	14.8	0.0

Totals			**707.5**						**707.5**	

Air System Heat Balance Summary for DOAS-1-CORRIDORS ONLY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) — 05/29/2026

Prepared by: Shakespeare Engineering — 1:25 PM

Table 1. System Loads

	DESIGN COOLING - JULY 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
COMPONENT LOADS	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Zone Conditioning	-	6623	937	-	3580	0
Plenum Load	-	0	0	-	0	0
Exhaust Fan Load	408 CFM	0	-	408 CFM	0	-
Ventilation Load	708 CFM	14483	-935	708 CFM	30612	0
Ventilation Fan Load	708 CFM	1049	-	708 CFM	-1049	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	**-**	**22155**	**2**	**-**	**33143**	**0**
Cooling Coil	-	22691	0	-	0	0
Heating Coil	-	0	-	-	33282	-
>> Total Conditioning	**-**	**22691**	**0**	**-**	**33282**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Table 2. Zone Heat Balance Loads

	DESIGN COOLING - JULY 15:00			DESIGN HEATING		
	OA DB / WB 106.2 F / 66.2 F			OA DB / WB 20.2 F / 16.3 F		
Zone Heat Balance Component	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**	**Details**	**Sensible [BTU/hr]**	**Latent [BTU/hr]**
Exterior Wall Convection	1146 sqft	1967	-	1146 sqft	4048	-
Roof Convection	0 sqft	0	-	0 sqft	0	-
Window Convection	528 sqft	3390	-	528 sqft	10039	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	0 sqft	0	-	0 sqft	0	-
Floor Convection	12124 sqft	8518	-	12124 sqft	3822	-
Interior Wall Convection	31821 sqft	14595	-	31821 sqft	2181	-
Ceiling Convection	12137 sqft	13259	-	12137 sqft	7172	-
Overhead Lighting Convection	3513 W	5322	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	2494 W	6383	-	0 W	0	-
People Convection	5	361	1007	0	0	0
Infiltration	0 CFM	0	0	0 CFM	0	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0		-	0	0
Safety Factor	0% / 0%	0	0	0%	0	0
>> Total Zone Loads	**-**	**53794**	**1007**	**-**	**27263**	**0**
Key:	**Positive values are cooling loads Negative values are heating loads**			**Positive values are heating loads Negative values are cooling loads**		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.